

机器视觉与 图像处理

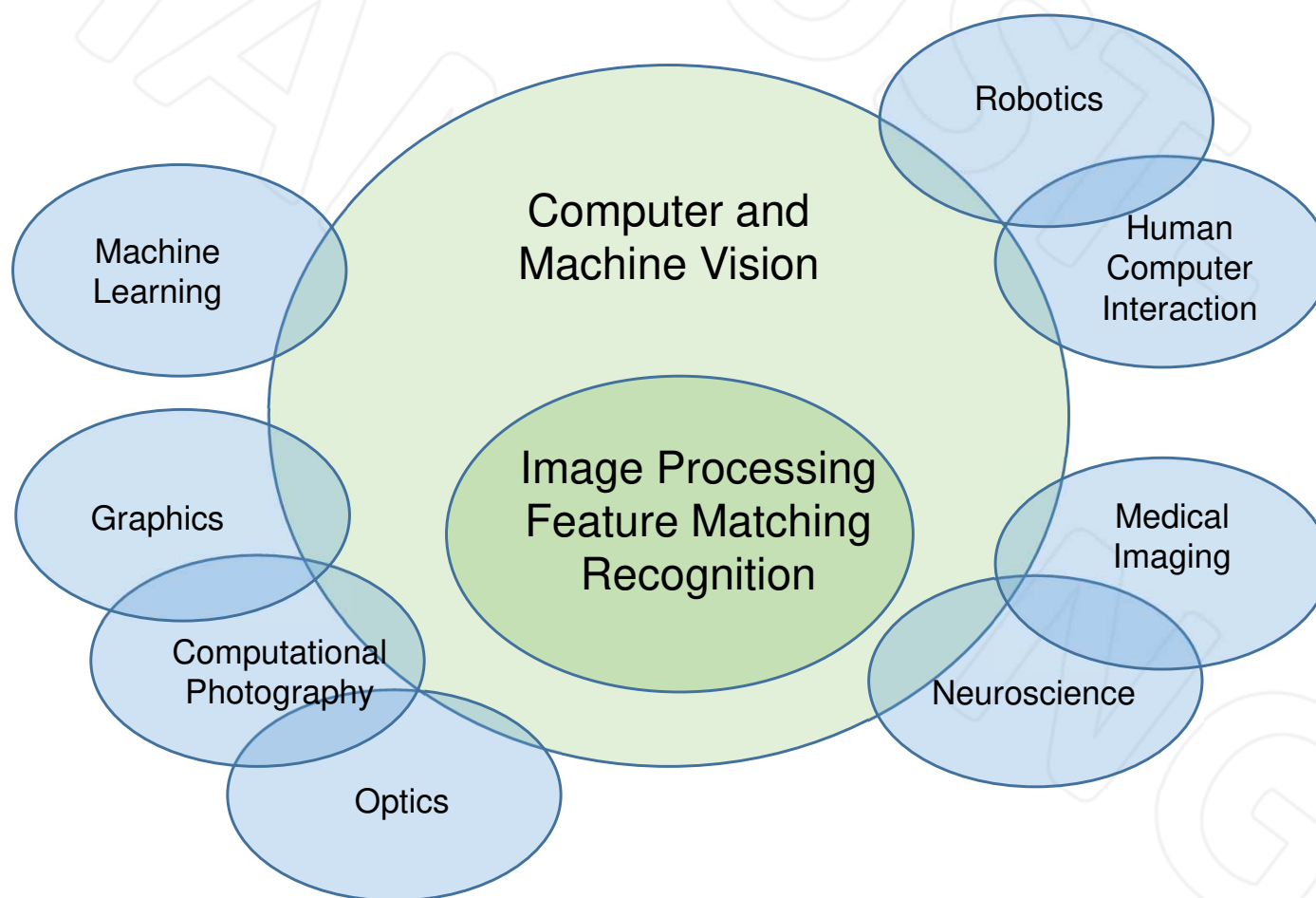
Machine Vision & Image Processing

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前次课程：机器视觉与图像处理绪论

- 机器视觉，视觉理论发展，机器视觉应用实例
- 图像处理概念，技术发展及应用



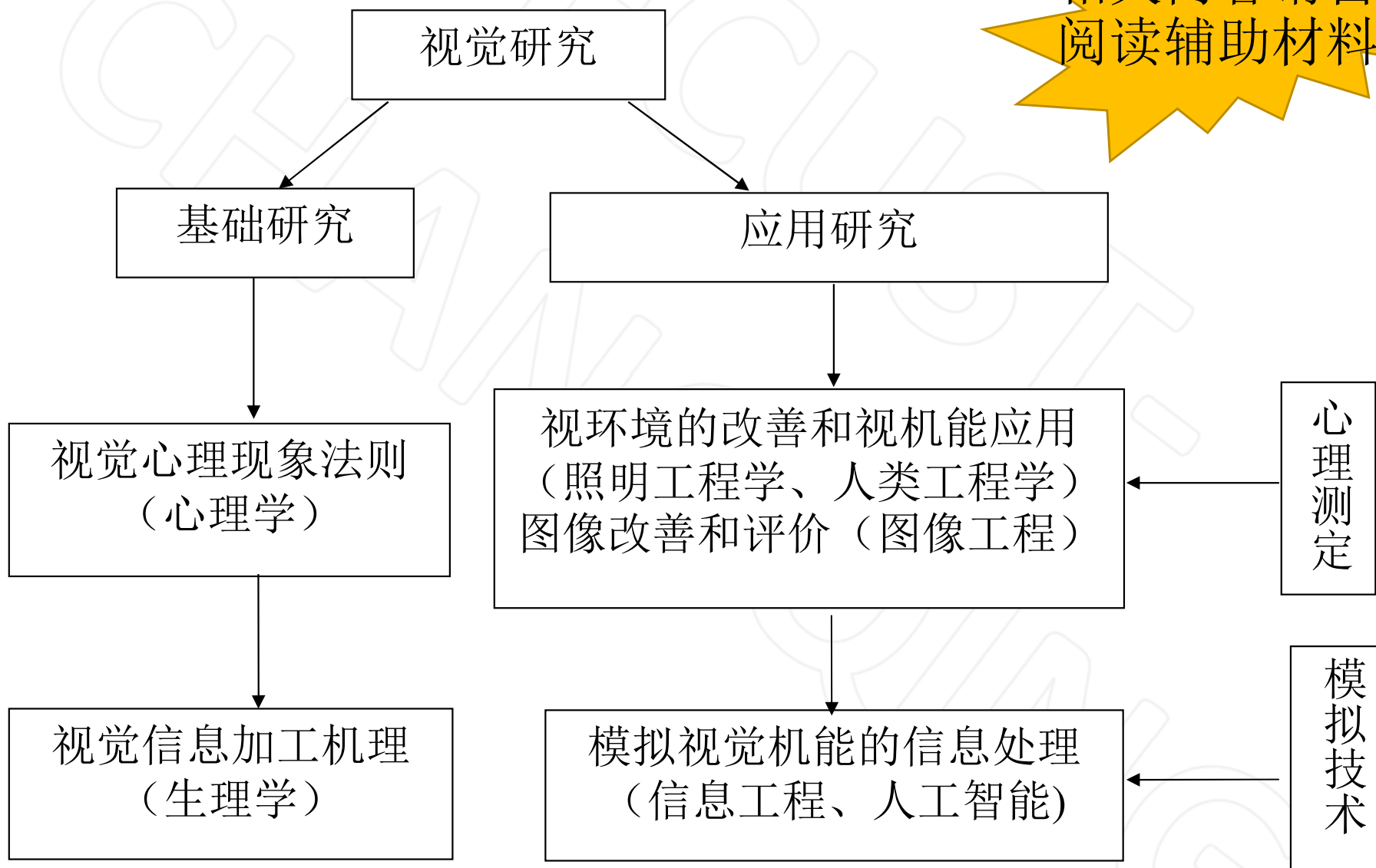
本次课程内容

- 视觉研究与图像的关系
 - 人眼结构与视觉特性
 - 有趣的视觉现象
- 投影几何与相机模型
 - 图像坐标和世界坐标的映射
 - 针孔相机模型
 - 投影几何(projective geometry)
 - 灭点(vanishing points)和灭线(vanishing lines)
 - 投影矩阵

图像处理的视觉基础

视觉研究与图像的关系

相关内容请自行
阅读辅助材料



A camera creates an image ...



world

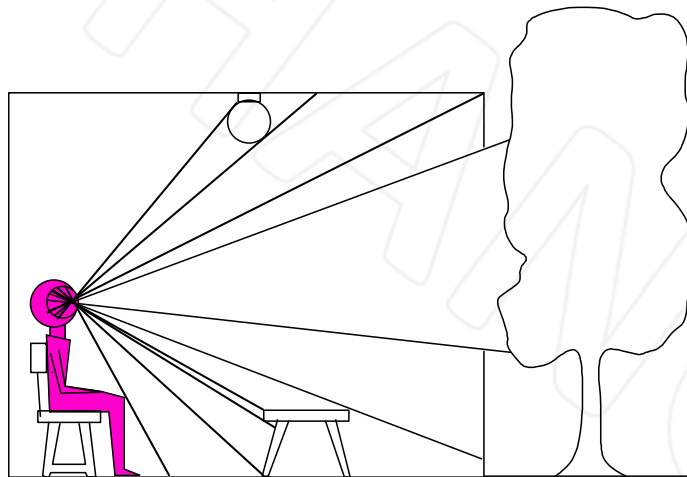
The image $I(x,y)$ measures how much light is captured at pixel (x,y)

We want to know

- Where does a point (X,Y,Z) in the world get imaged?
- What is the brightness at the resulting point (x,y) ?

Dimensionality Reduction Machine (3D to 2D)

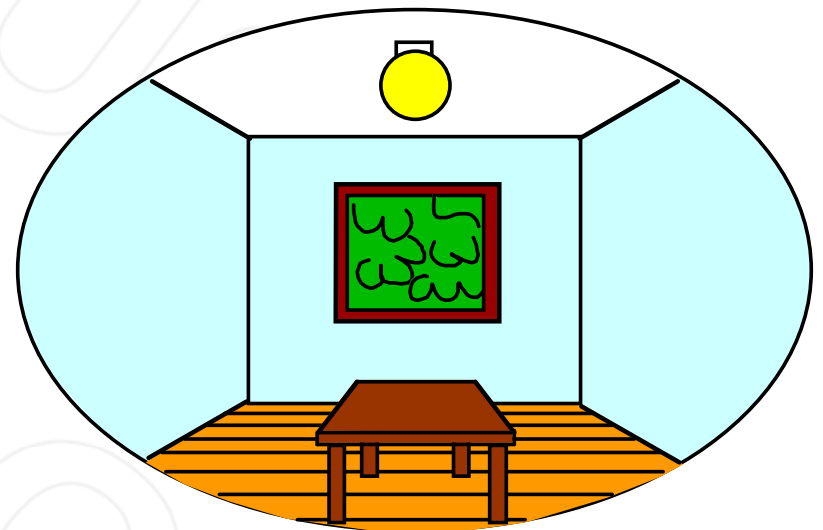
3D world



Point of observation



2D image

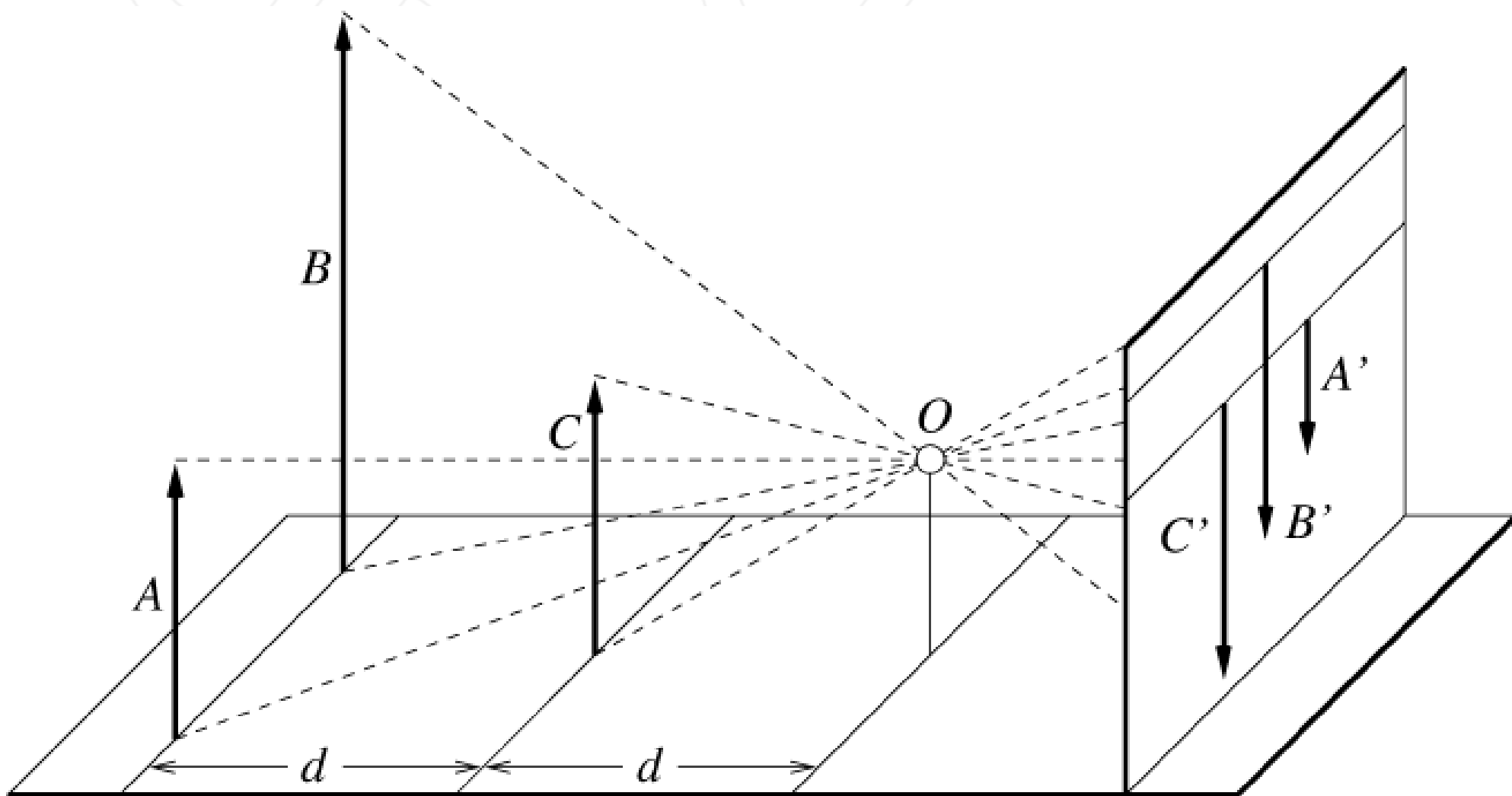


Intrinsic ambiguity of the mapping from 3D to image (2D)



投影几何与相机模型

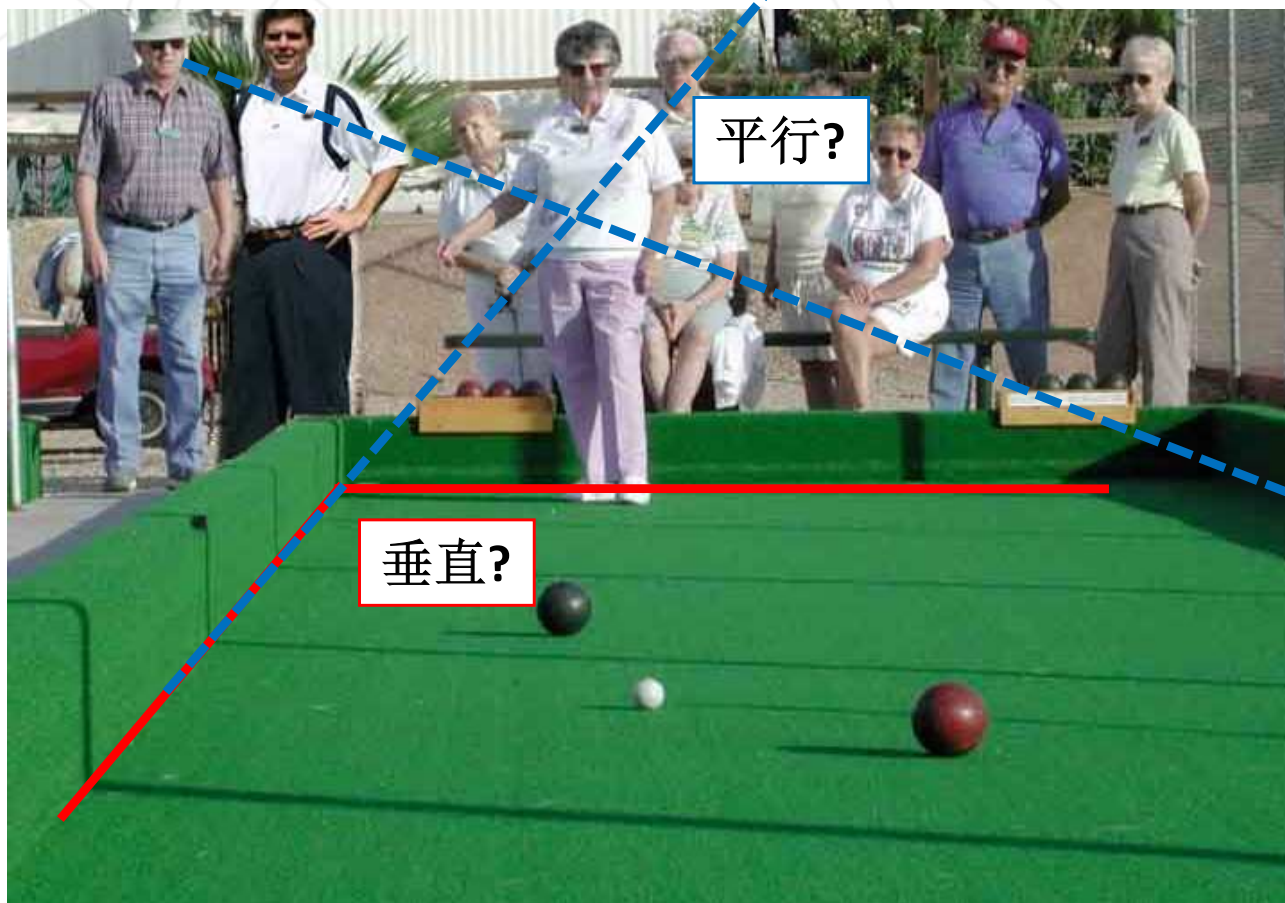
- 长度没有保持



投影几何

丢失了什么信息？

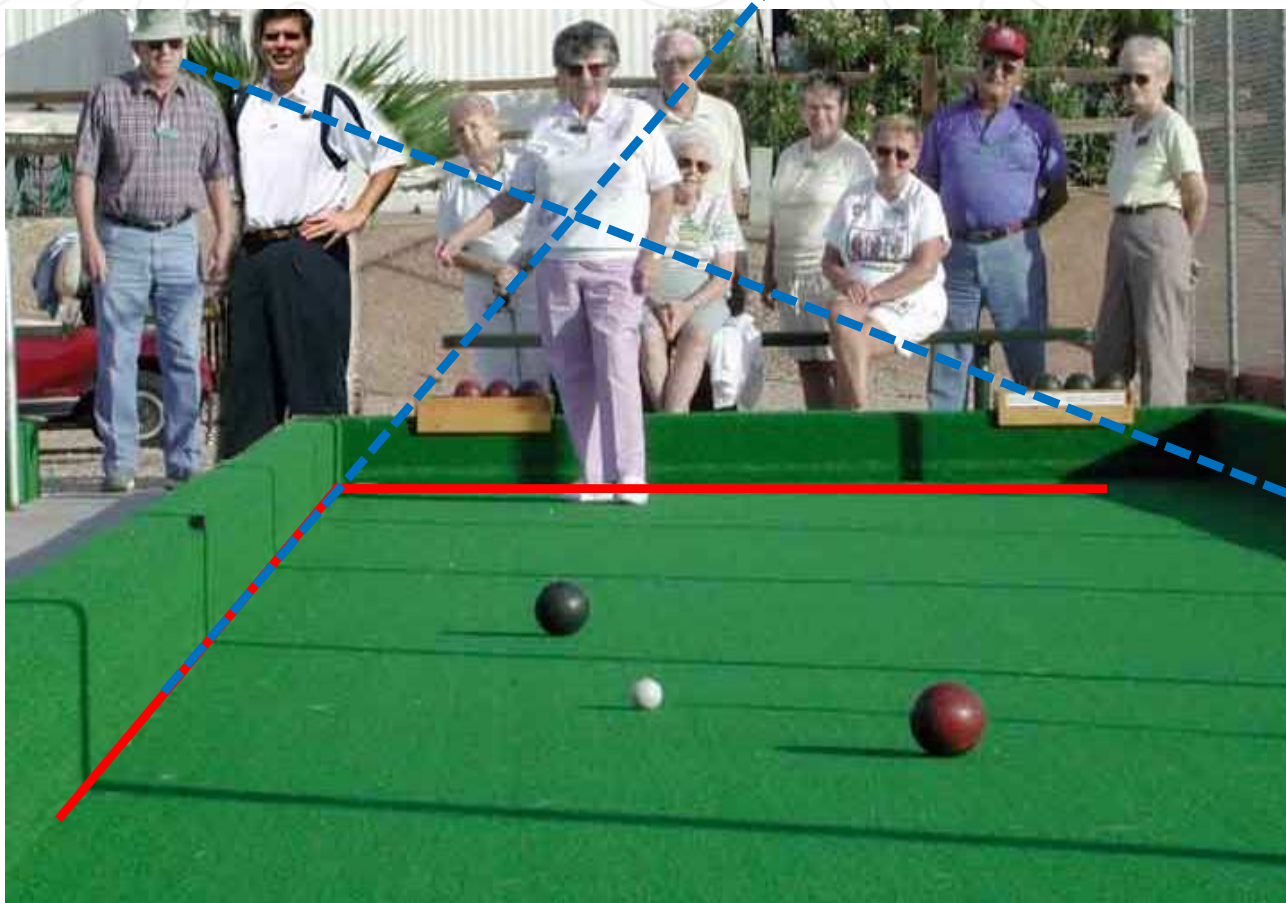
- 长度信息
- 角度信息



投影几何

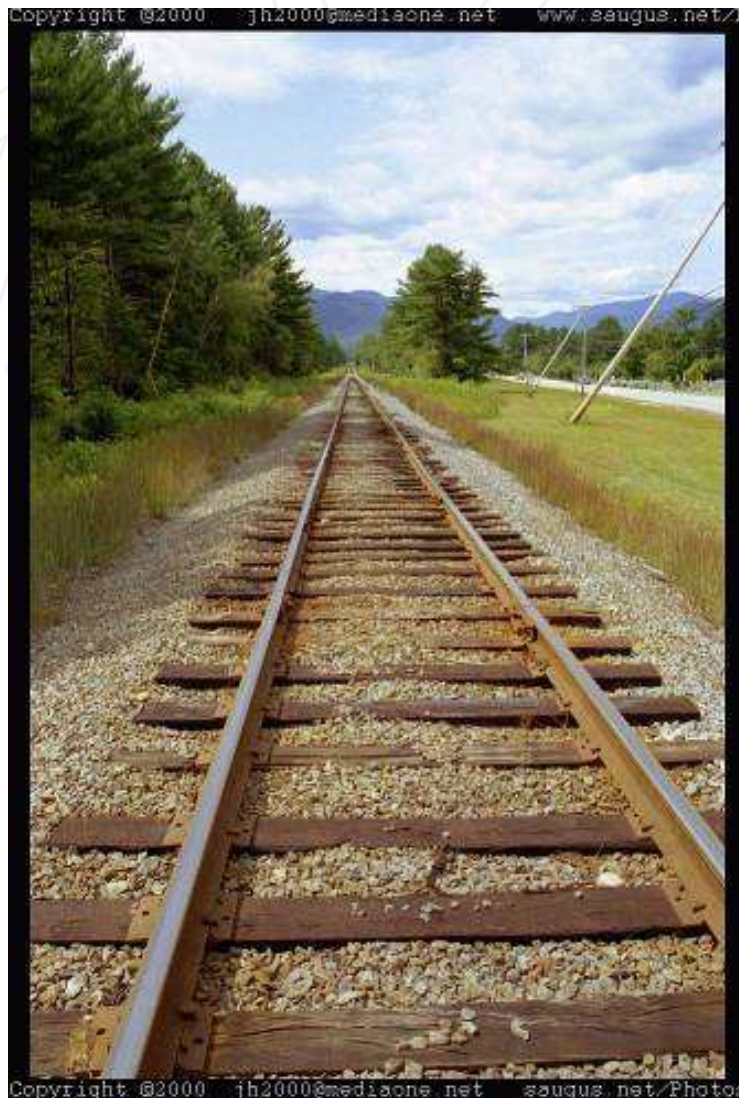
保持了什么信息？

- 直线依然是直的

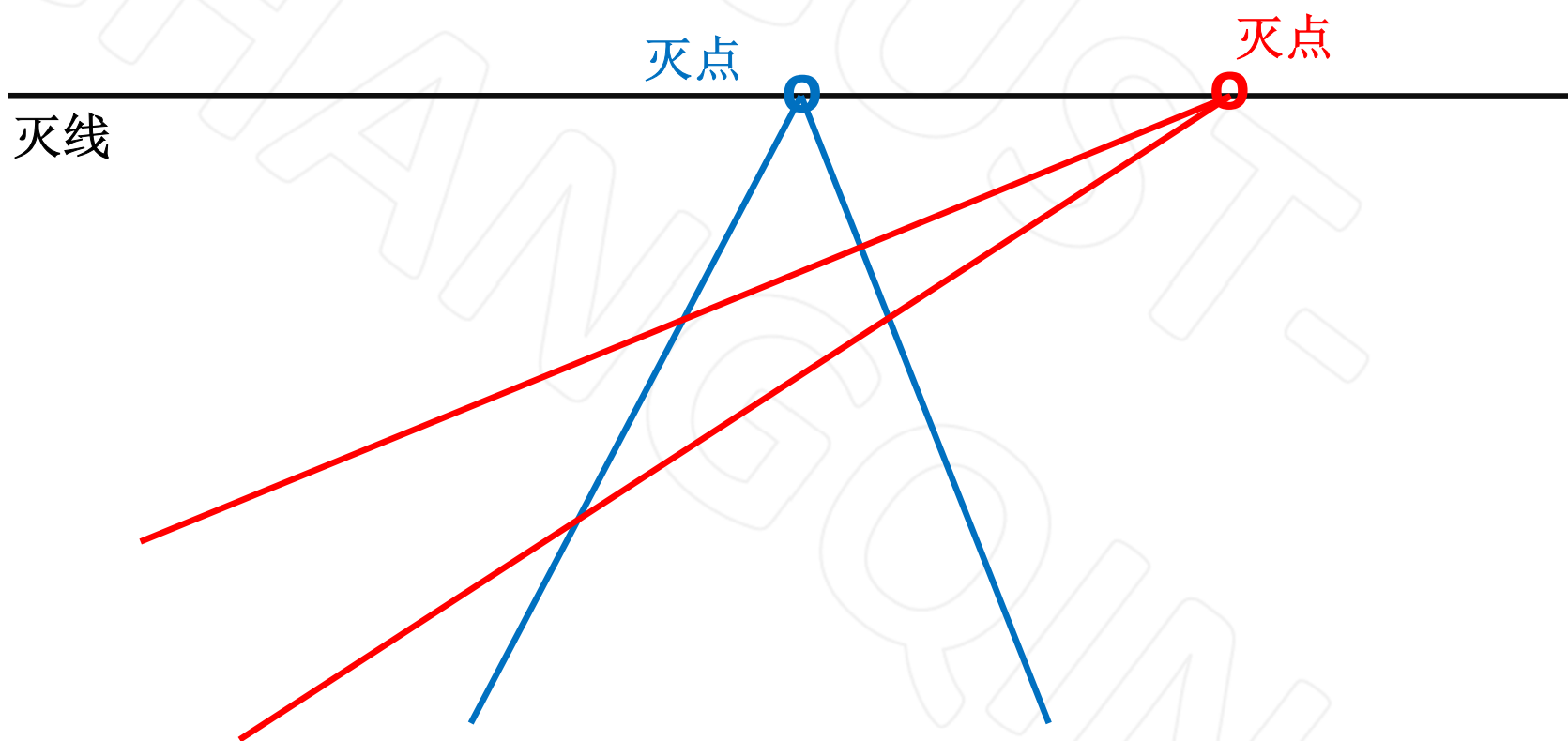


灭点(vanishing points)和灭线(vanishing lines)

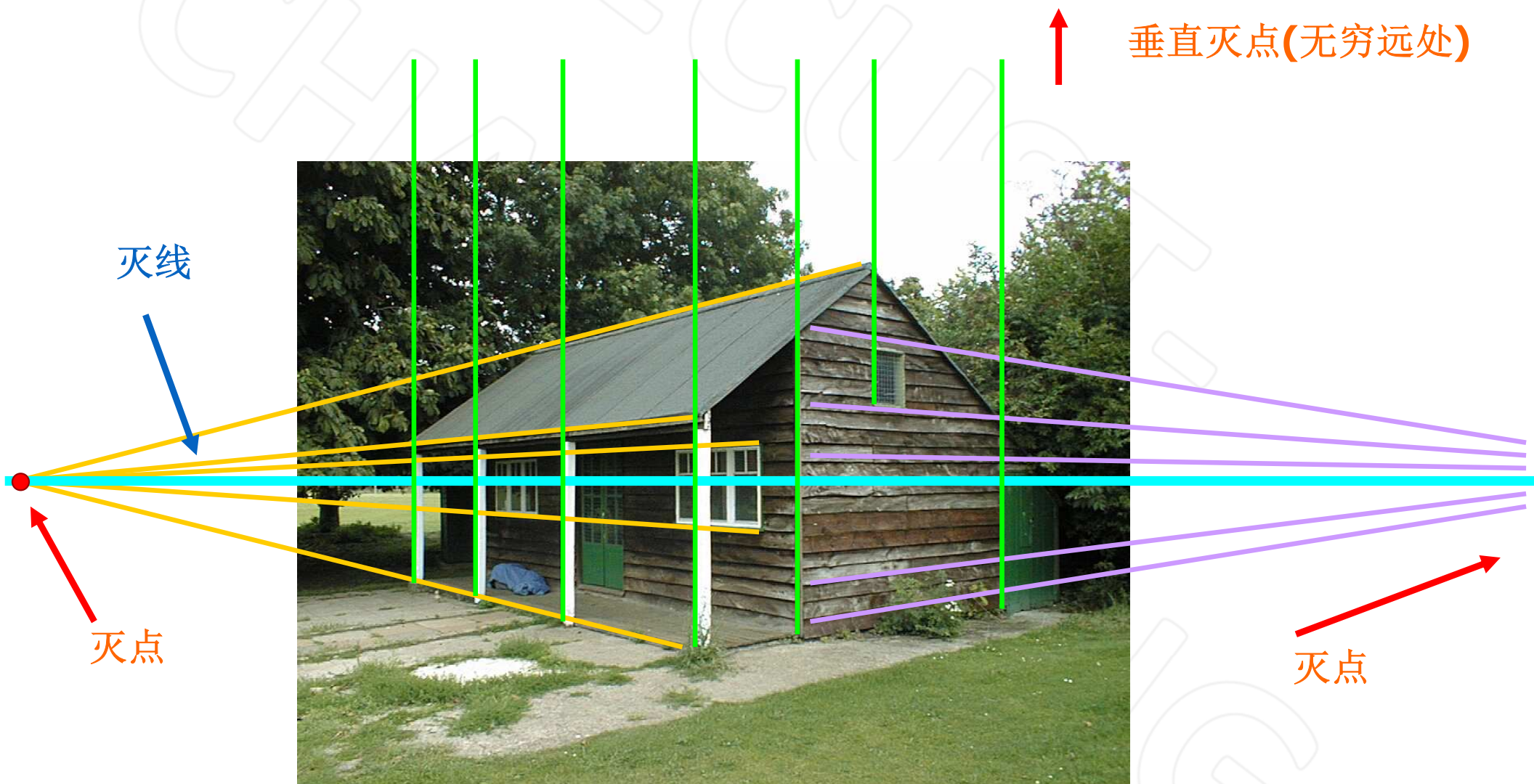
现实世界中的平行线在图像中交会于“灭点”



灭点(vanishing points)和灭线(vanishing lines)



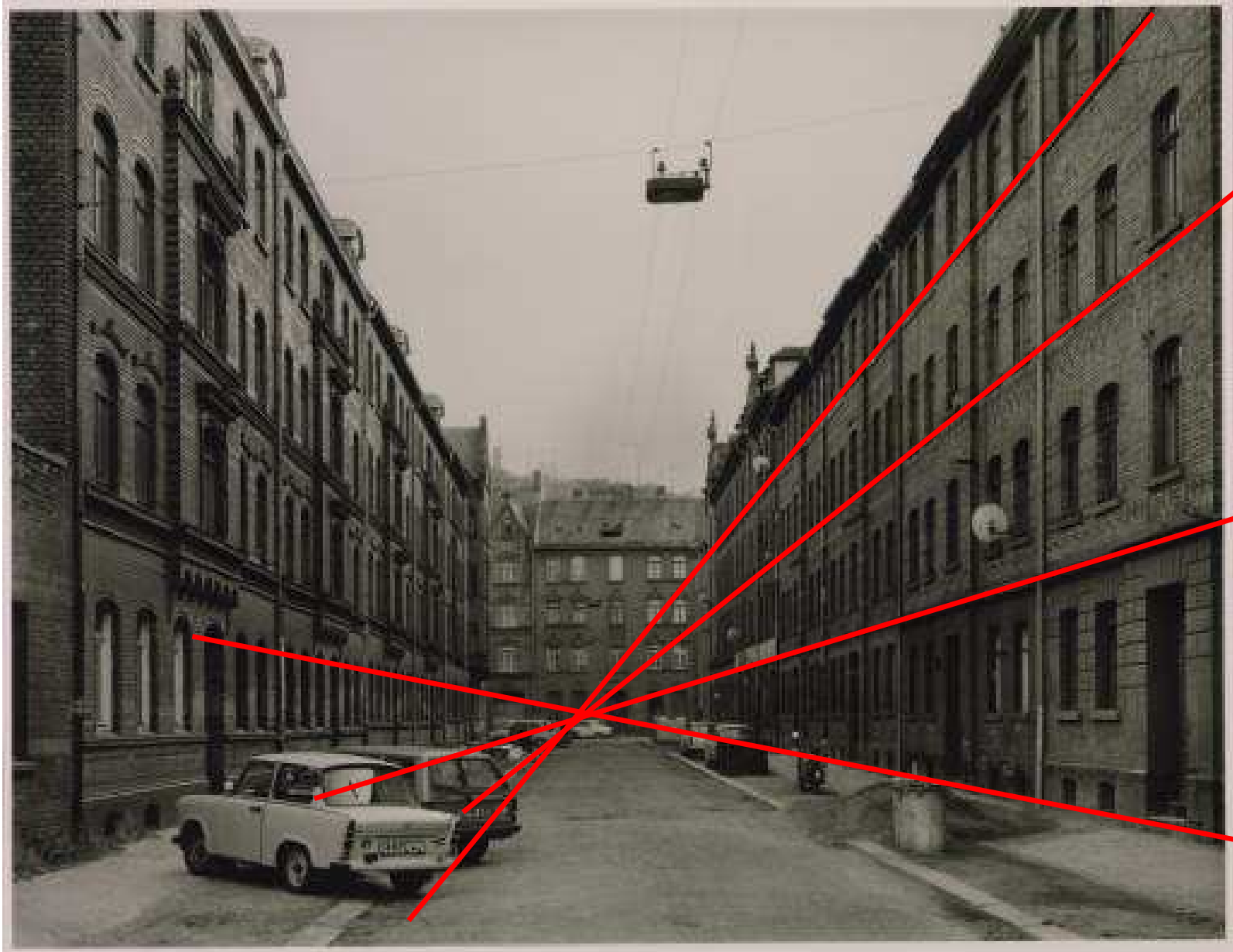
灭点(vanishing points)和灭线(vanishing lines)



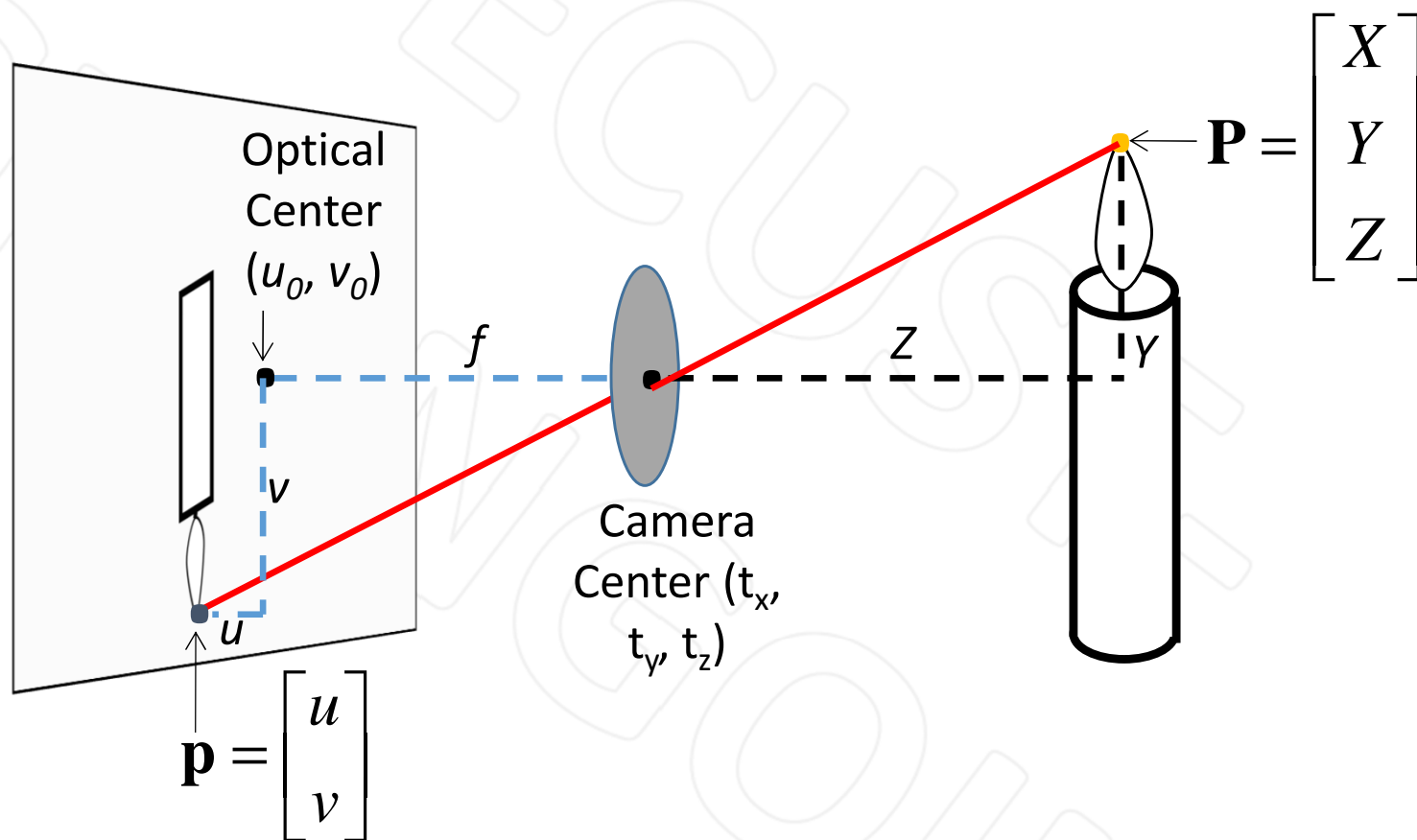
灭点(vanishing points)和灭线(vanishing lines)



灭点(vanishing points)和灭线(vanishing lines)

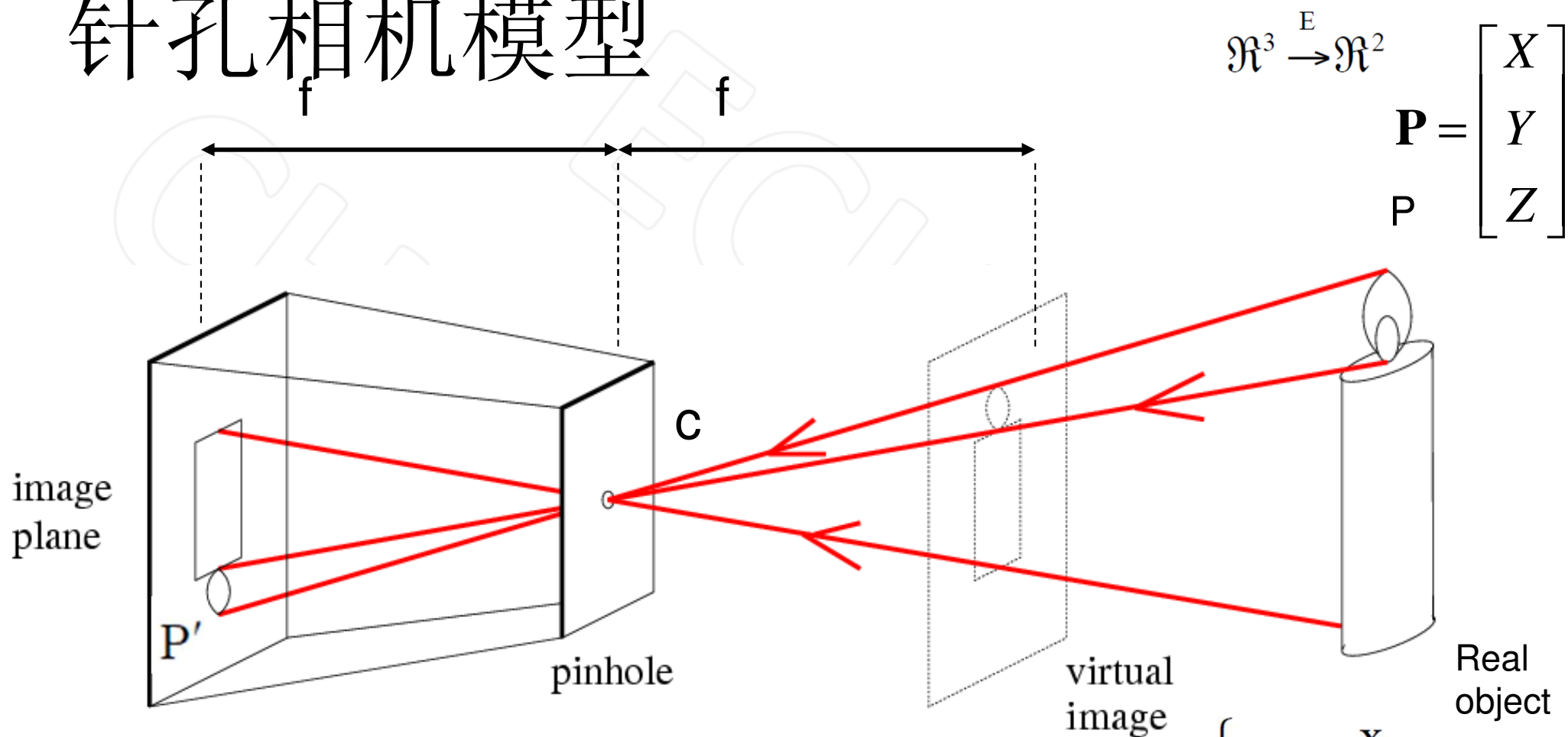


映射：世界坐标→图像坐标



世界坐标系(world coordinate system)
相机坐标系(camera coordinate system)
图像坐标系(image coordinate system)
像素坐标系(pixel coordinate system)

针孔相机模型

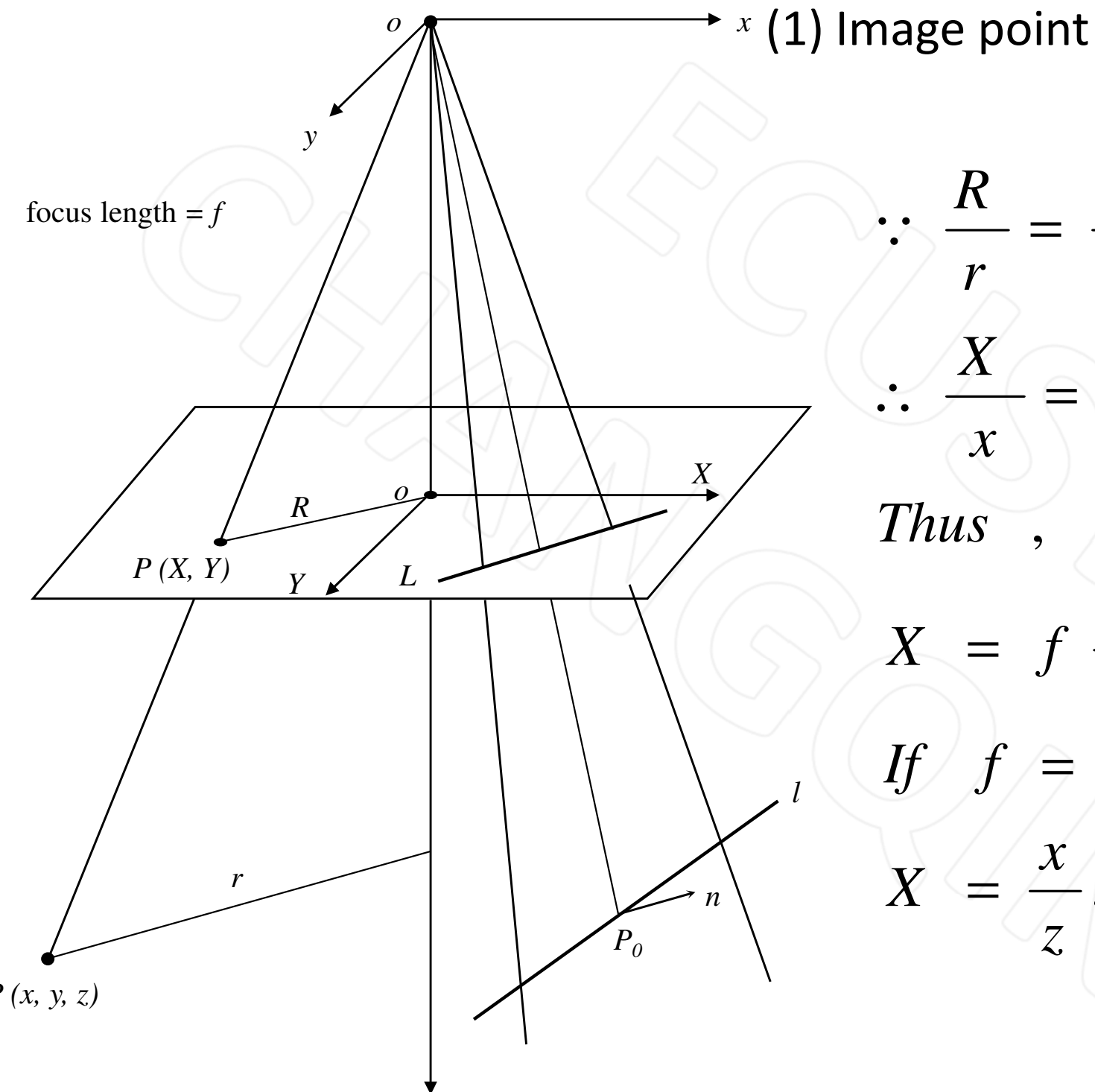


f = Focal length

c = Optical center of the camera

$$P = (x, y, z) \rightarrow P' = \left(f \frac{x}{z}, f \frac{y}{z} \right)$$

$$\begin{cases} x' = f \frac{x}{z} \\ y' = f \frac{y}{z} \end{cases}$$



$$\therefore \frac{R}{r} = \frac{f}{z}, \quad \frac{X}{x} = \frac{Y}{y} = \frac{R}{r}$$

$$\therefore \frac{X}{x} = \frac{Y}{y} = \frac{f}{z}$$

Thus ,

$$X = f \frac{x}{z}, \quad Y = f \frac{y}{z}$$

If $f = 1$,

$$X = \frac{x}{z}, \quad Y = \frac{y}{z}$$

Is this a linear transformation?

$$P = (x, y, z) \rightarrow P' = \left(f \frac{x}{z}, f \frac{y}{z}\right)$$

No –division by z is nonlinear!

How to make it linear?

齐次坐标系(Homogeneous coordinates)

- 变换 (conversion)

普通坐标变换为齐次坐标

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

齐次图像坐标

$$(x, y, z) \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

齐次世界坐标

齐次坐标变换

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

2D (image) coordinates

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \Rightarrow (x/w, y/w, z/w)$$

3D (scene) coordinates

齐次坐标系(Homogeneous coordinates)

- 伸缩不变性

$$k \begin{bmatrix} x \\ y \\ w \end{bmatrix} = \begin{bmatrix} kx \\ ky \\ kw \end{bmatrix} \Rightarrow \begin{bmatrix} \frac{kx}{kw} \\ \frac{ky}{kw} \\ \frac{kw}{kw} \end{bmatrix} = \begin{bmatrix} \frac{x}{w} \\ \frac{y}{w} \\ 1 \end{bmatrix}$$

齐次坐标
Homogeneous
Coordinates

笛卡尔坐标
Cartesian Coordinates

笛卡尔坐标系中的点在齐次坐标系中是一条直线

齐次坐标系：几何基础

- 直线方程: $ax + by + c = 0$

$$line_i = \begin{bmatrix} a_i \\ b_i \\ c_i \end{bmatrix}$$

- 在像素坐标后填1就得到齐次坐标

$$p_i = \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix}$$

- 直线可以由两点的向量积(cross product)获得

$$line_{ij} = p_i \times p_j$$

- 两条直线的交点由这两条直线的向量积获得

$$q_{ij} = line_i \times line_j$$

齐次坐标系：平行线的交点？

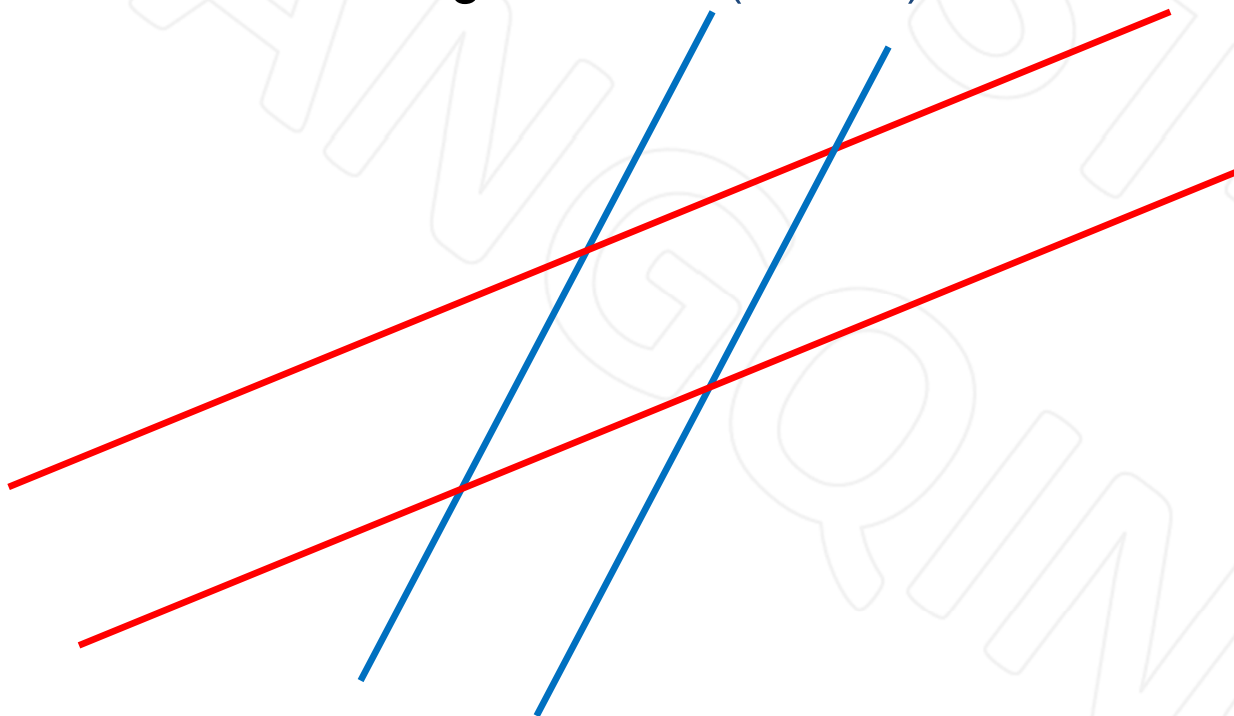
- 平行线相交于不同的无穷远点

Cartesian: (Inf, Inf)

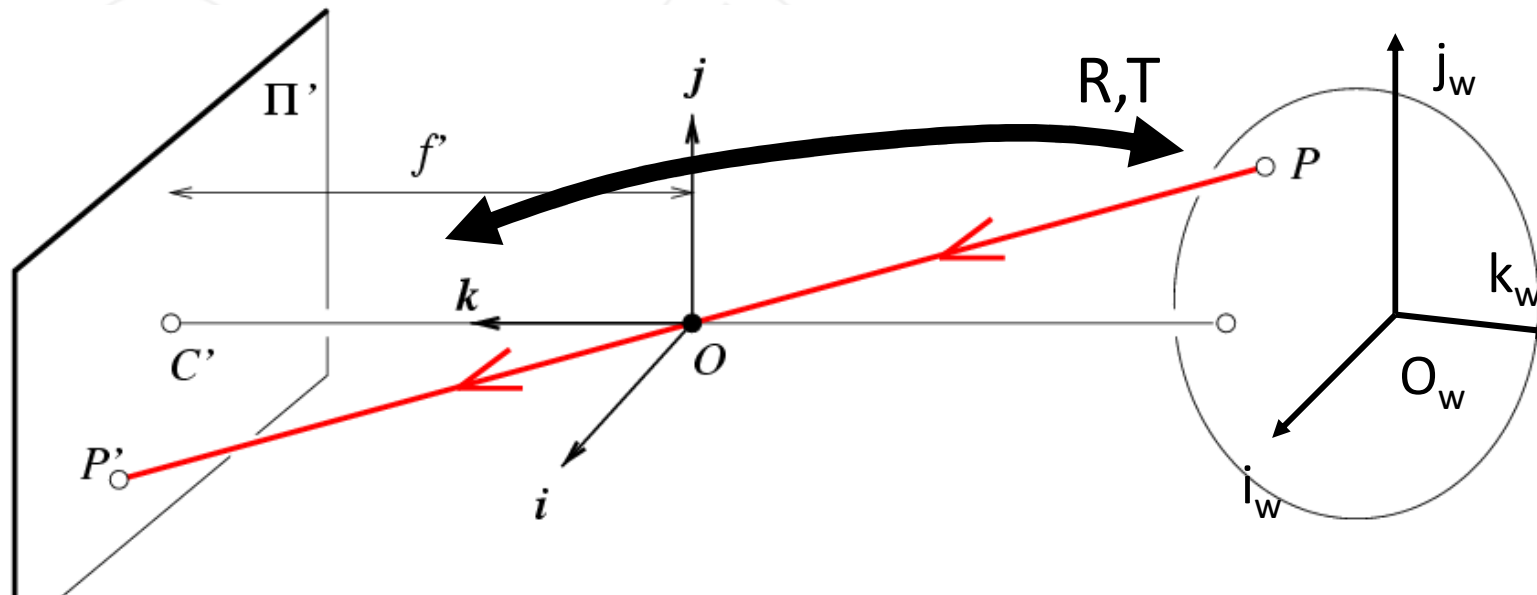
Homogeneous: $(1, 1, 0)$

Cartesian: (Inf, Inf)

Homogeneous: $(1, 2, 0)$



投影矩阵 (Projection matrix)



Intrinsic Assumptions

- Optical center at (u_0, v_0)
- Rectangular pixels
- Small skew

Extrinsic Assumptions

- Allow rotation
- Camera at (t_x, t_y, t_z)

$$P' = K [R \quad \bar{t}] P \Rightarrow {}^w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \alpha & s & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}}_M \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

5 6

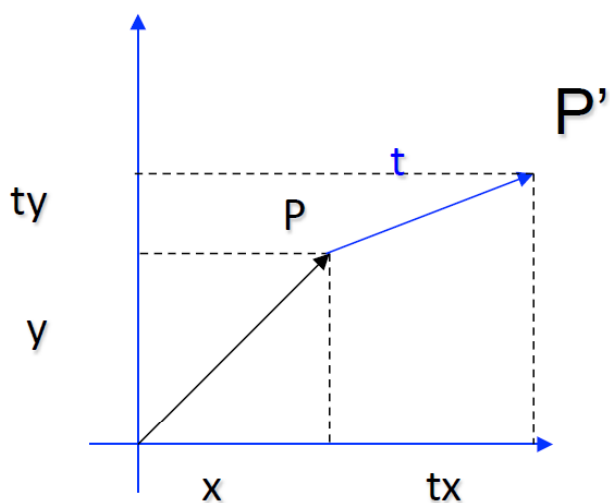
“Projection matrix”

↓

$P' = M P$

投影矩阵:

2D Translation using Homogeneous Coordinates



$$\mathbf{P} = (x, y) \rightarrow (x, y, 1)$$

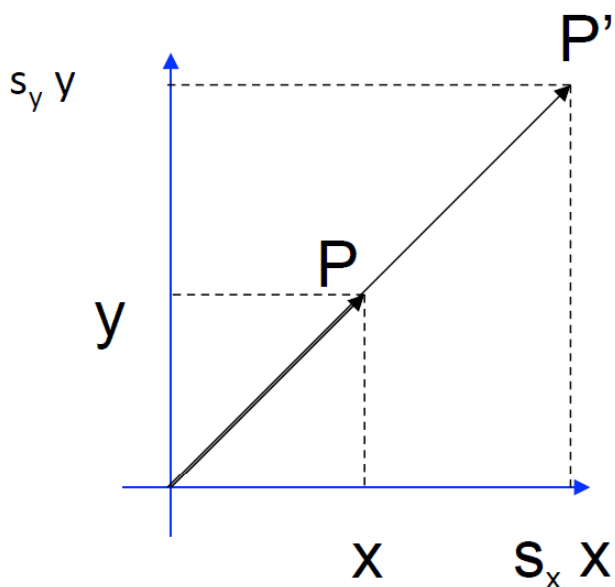
$$\mathbf{t} = (t_x, t_y) \rightarrow (t_x, t_y, 1)$$

$$\begin{aligned} \mathbf{P}' &\rightarrow \begin{bmatrix} x + t_x \\ y + t_y \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{I} & \mathbf{t} \\ 0 & 1 \end{bmatrix} \cdot \mathbf{P} = \mathbf{T} \cdot \mathbf{P} \end{aligned}$$

The diagram shows the matrix multiplication for 2D translation. The first matrix is a 3x3 matrix with rows [1, 0, t_x], [0, 1, t_y], and [0, 0, 1]. The second matrix is a 3x1 column vector [x, y, 1]. The result is a 3x1 column vector [x + t_x, y + t_y, 1]. The matrix is also written as [I t; 0 1] multiplied by P, which equals T multiplied by P. Arrows point from the labels t and P in the equations above to the corresponding parts of the matrix multiplication.

投影矩阵：

Scaling Equation



$$\mathbf{P} = (x, y) \rightarrow \mathbf{P}' = (s_x x, s_y y)$$

$$\mathbf{P} = (x, y) \rightarrow (x, y, 1)$$

$$\mathbf{P}' = (s_x x, s_y y) \rightarrow (s_x x, s_y y, 1)$$

$$\mathbf{P}' \rightarrow \begin{bmatrix} s_x x \\ s_y y \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{S}} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{S}' & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{bmatrix} \cdot \mathbf{P} = \mathbf{S} \cdot \mathbf{P}$$

投影矩阵：

Scaling & Translating

$$\begin{aligned}\mathbf{P}'' &= \mathbf{T} \cdot \mathbf{S} \cdot \mathbf{P} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \\ &= \underbrace{\begin{bmatrix} s_x & 0 & t_x \\ 0 & s_y & t_y \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} s_x x + t_x \\ s_y y + t_y \\ 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} S & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}\end{aligned}$$

投影矩阵:

Translating & Scaling
!= Scaling & Translating

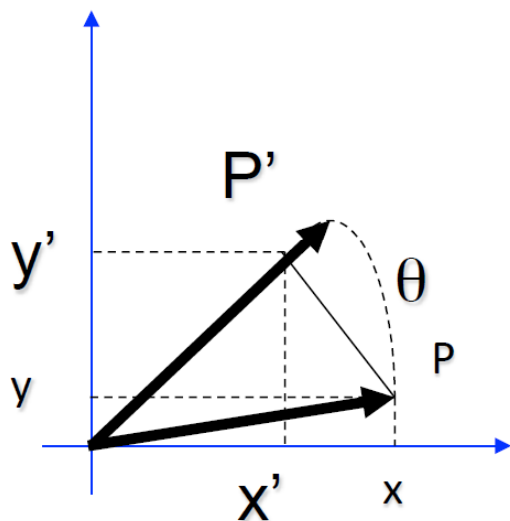
$$\mathbf{P}''' = \mathbf{T} \cdot \mathbf{S} \cdot \mathbf{P} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & t_x \\ 0 & s_y & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} s_x x + t_x \\ s_y y + t_y \\ 1 \end{bmatrix}$$

$$\begin{aligned} \mathbf{P}''' = \mathbf{S} \cdot \mathbf{T} \cdot \mathbf{P} &= \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \\ &= \begin{bmatrix} s_x & 0 & s_x t_x \\ 0 & s_y & s_y t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} s_x x + s_x t_x \\ s_y y + s_y t_y \\ 1 \end{bmatrix} \end{aligned}$$

投影矩阵：

Rotation Equations

Counter-clockwise rotation by an angle θ



$$x' = \cos \theta \, x - \sin \theta \, y$$

$$y' = \cos \theta \, y + \sin \theta \, x$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{R} \, \mathbf{P}$$

投影矩阵：

Rotation Matrix Properties

- Transpose of a rotation matrix produces a rotation in the opposite direction

$$\mathbf{R} \cdot \mathbf{R}^T = \mathbf{R}^T \cdot \mathbf{R} = \mathbf{I}$$

$$\det(\mathbf{R}) = 1$$

- The rows of a rotation matrix are always mutually perpendicular (a.k.a. orthogonal) unit vectors
 - (and so are its columns)

投影矩阵:

Properties

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

A 2D rotation matrix is 2x2

Note: $\mathbf{R}$ belongs to the category of *normal* matrices and satisfies many interesting properties:

$$\mathbf{R} \cdot \mathbf{R}^T = \mathbf{R}^T \cdot \mathbf{R} = \mathbf{I}$$

$$\det(\mathbf{R}) = 1$$

投影矩阵:

Scaling + Rotation + Translation

$$P' = (T R S) P$$

$$P' = T \cdot R \cdot S \cdot P = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} =$$

$$= \begin{bmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} =$$

This is the form of the
general-purpose
transformation matrix

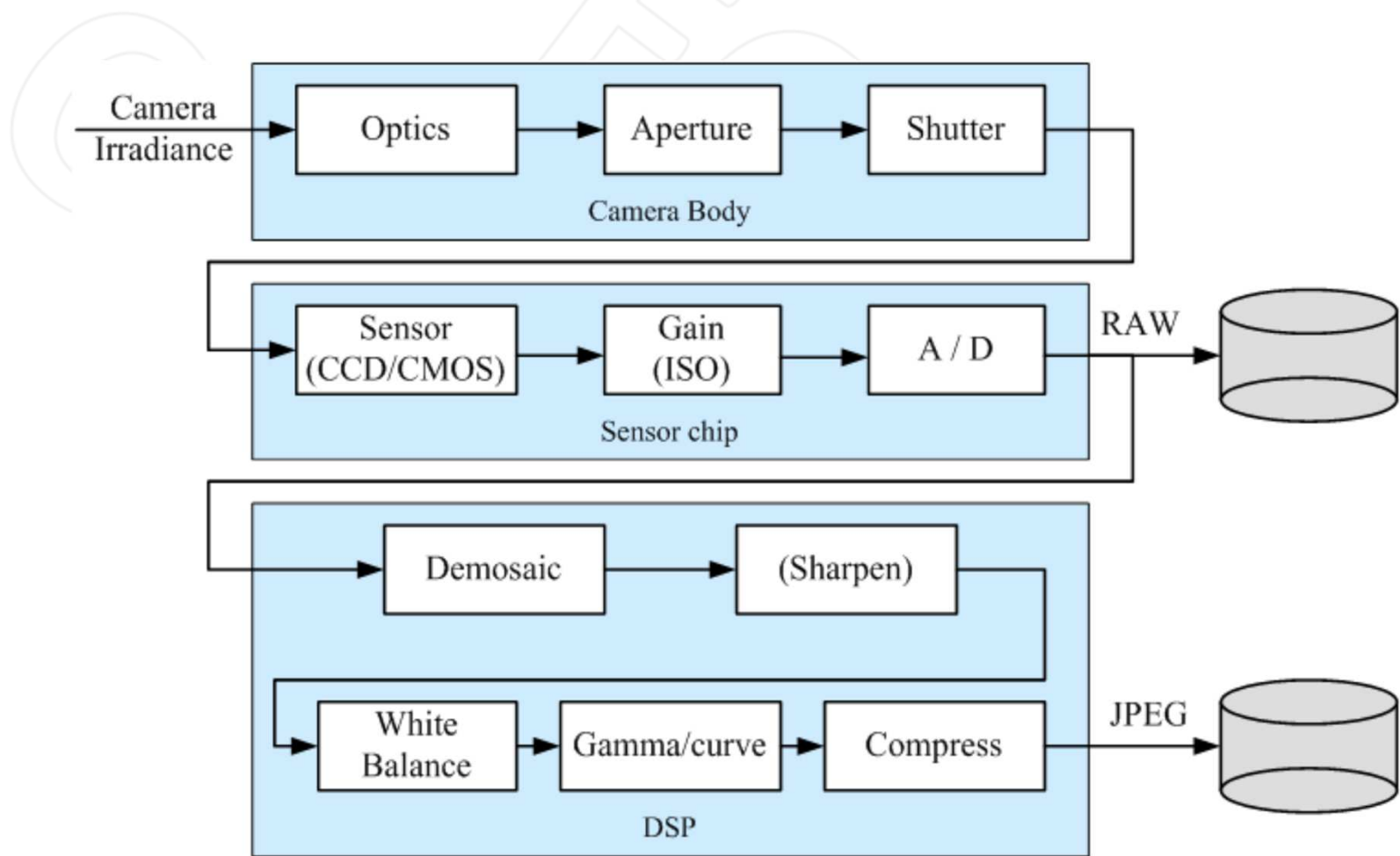
$$= \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} S & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \boxed{\begin{bmatrix} R S & t \\ 0 & 1 \end{bmatrix}} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

数码相机（Digital camera）



- 数码相机中，使用传感器阵列替代了胶片
 - 阵列中的每一个单元都是一个光敏二极管，将光信号转换为电信号
 - 两种常见的传感器类型：
 - **CCD**(Charge Coupled Device ,电荷耦合元件)
 - **CMOS**(Complementary Metal Oxide Semiconductor,互补金属氧化物半导体元件)

数码相机（Digital camera）



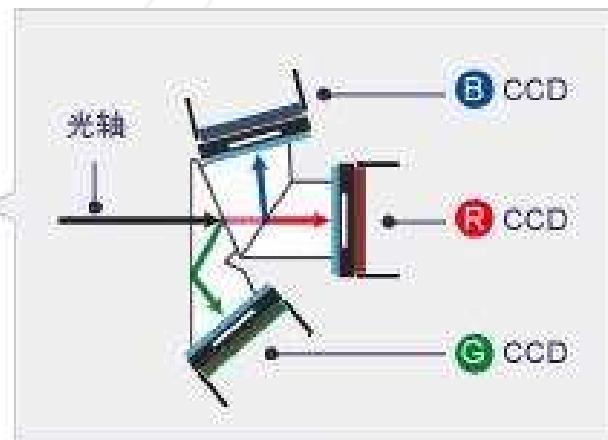
- 数码相机的成像的典型流程

- RAW格式存储未进行数字处理的图像
- JPEG存储根据相机设置处理后的压缩图像

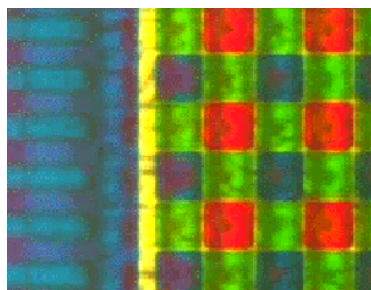
图像传感器

Image acquisition

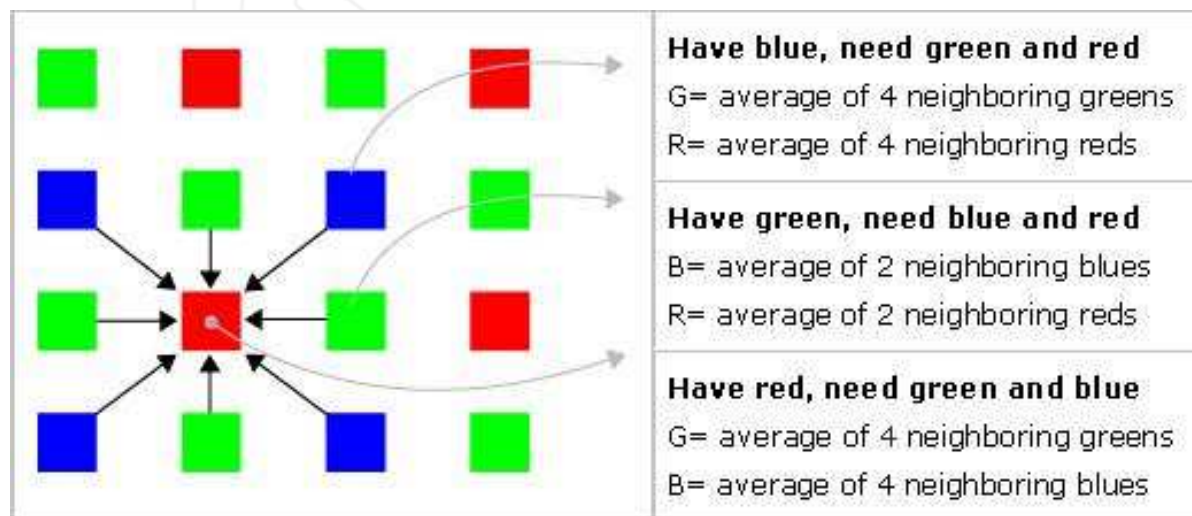
CCD Sensor



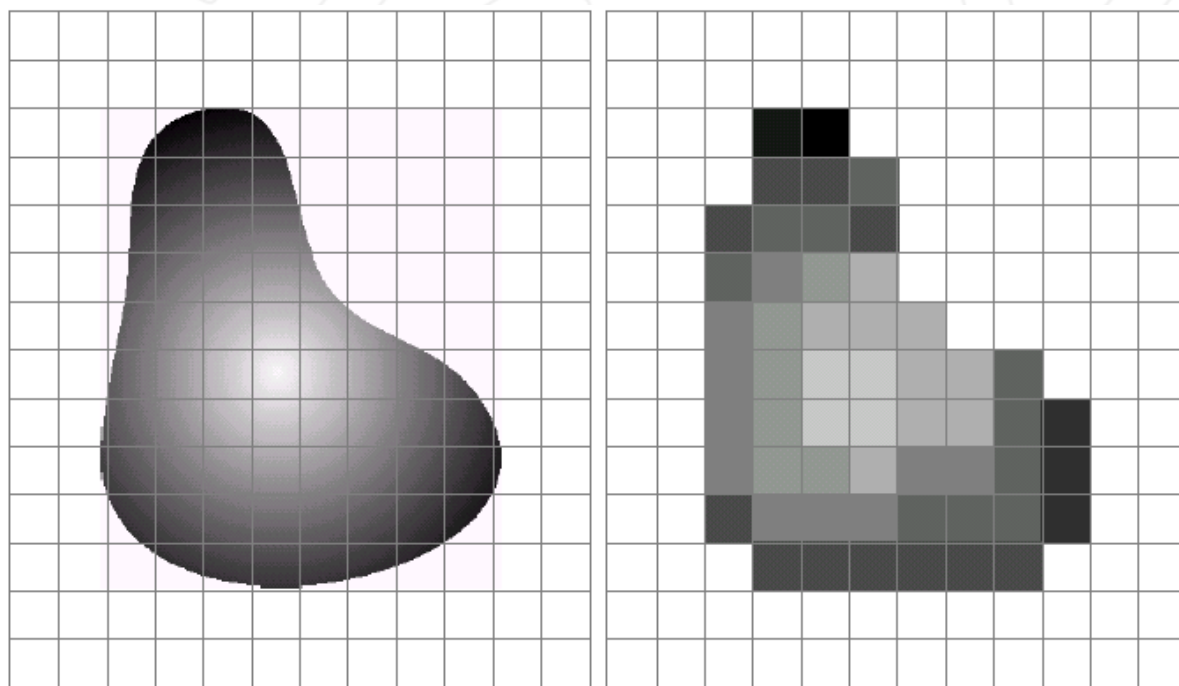
CMOS Sensor



HV7131E

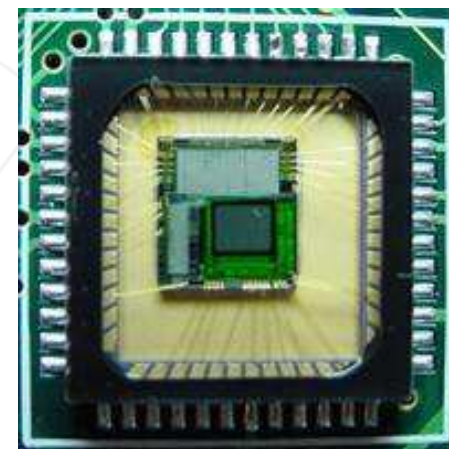


相机传感器阵列(Sensor Array)



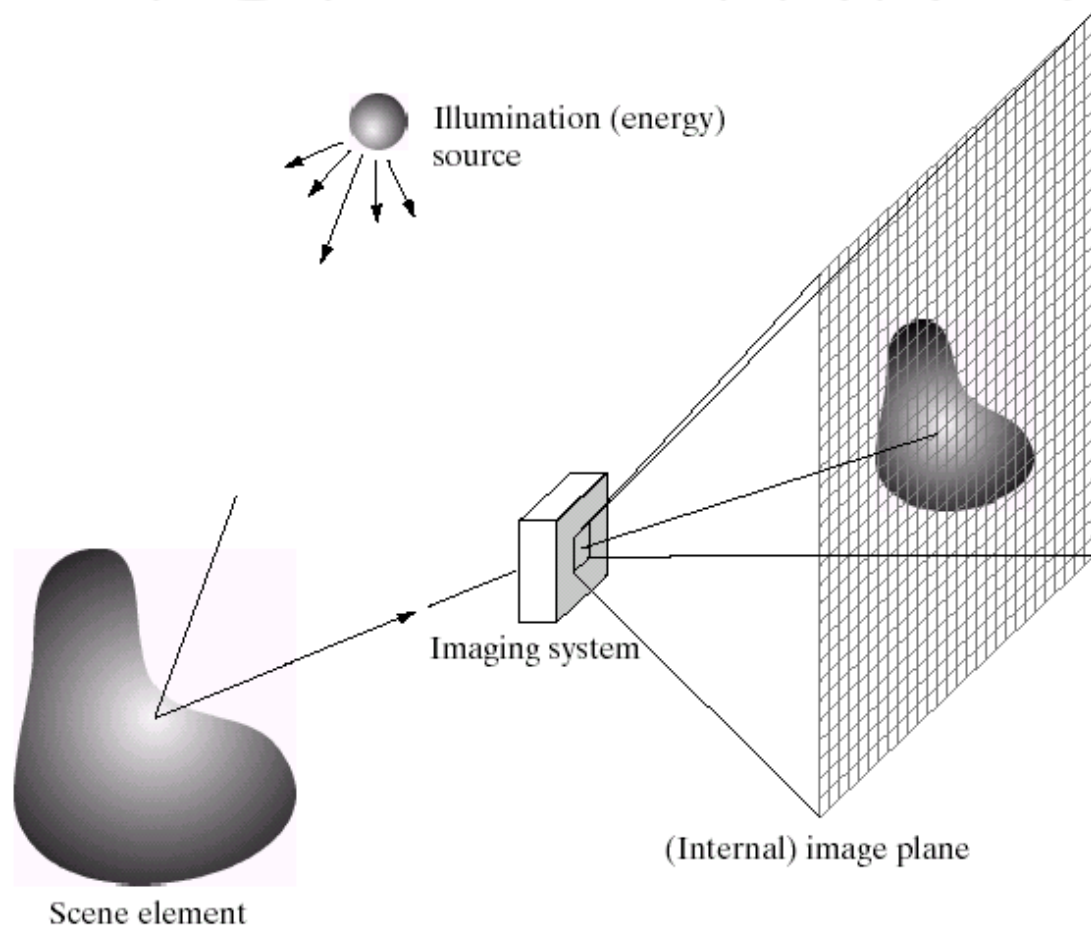
a b

FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

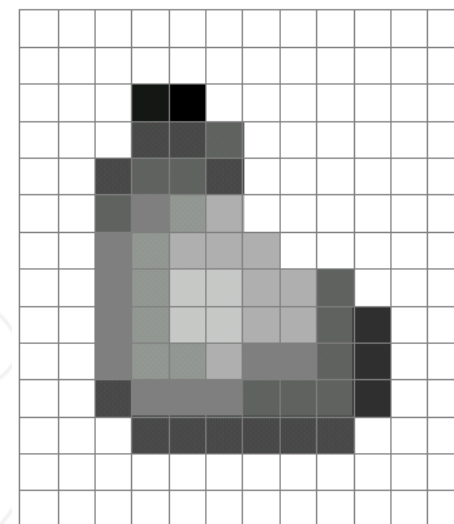


CMOS sensor

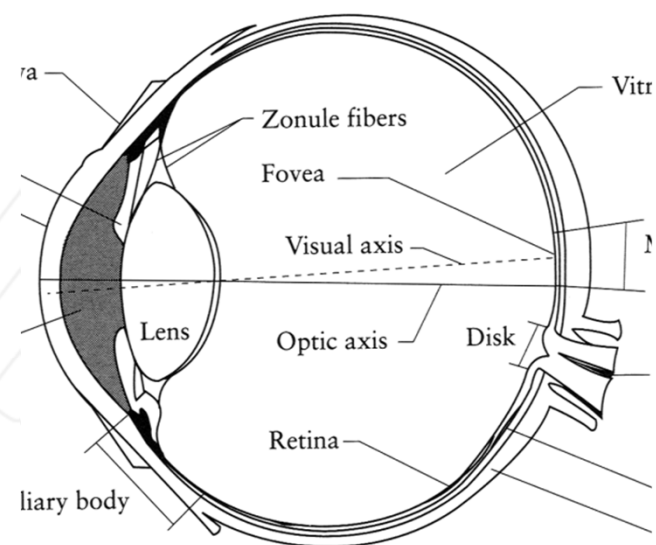
图像格式



胶片



图像格式



人眼